Amesp

Atomic and molecular electronic structure program

Version 2.1(dev)

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Program Manual

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1. Introduction

Atomic and molecular electronic structure program (Amesp) is a free *ab initio*, density functional and semiempirical methods quantum chemistry program. It has been independently developed by Zhang Yingfeng since 2016-03-09. Amesp supports many WFT, DFT and semiempirical methods and their first-order and second-order analytical derivatives. It supports excited state calculations, and can consider solvation and relativistic effects. It can perform geometric structure optimization, transition state search and vibration analysis. SCF adopts many methods to improve the stability and convergence, while using the methods such as RI and COSX to accelerate the calculation of energy and energy derivatives.

2. Authors

The main authors of the Amesp and their major contributions are as follows:

Zhang YingFeng :

(1) General design.

(2) All electronic integrals and integral derivatives.

(3) Initial Guess, A/E/C/QDIIS, SOSCF, QCSCF and Symmetry.

(4) Conventional and RICOSX based HF, DFT, MP2.

(5) Conventional MP3, MP4, CID, CISD, QCISD, CCD, CCSD, CCSD(T).

(6) Conventional CASSCF, CASCI and Full CI.

(7) Conventional and RICOSX based CIS, CIS(D), TDHF, TDDFT, TDA and SOC.

(8) Wavefunction stability analysis for R,U and RO.

(9) Atomic density based tight-binding method (aTB).

(10) Universal Force Field (UFF), Amber Force Field.

(11) First-order analysis derivative of HF, DFT, MP2, CIS, TDDFT, aTB, UFF and Amber Force Field.

(12) Second-order analysis derivative of HF, DFT, aTB, UFF and Amber Force Field.

(13) Exact two component (X2C) and first-order analysis derivative.

(14) Energy Decomposition Analysis (EDA).

(15) Optimization algorithm, including minimizations and saddle point optimizations.

(16) Intrinsic Reaction Coordinate (IRC).

(17) Ab Initio Molecular Dynamics (AIMD).

(18) Polarizable Continuum Model (PCM), Universal Easy Solvation Evaluation (uESE).

(19) Geometrical counterpoise (gCP).

(20) Some other methods and all auxiliary programs.

Zhao Jian :

(1) aesp programs.

Zhang Jintu :

(1) aopt programs : amesp's geometry optimization monitor.

Li Junhong :

(1) Gaussian-Amesp(gamesp.py) is a solution for invoking Amesp from Gaussian, which means one can do optimization, frequencies, or even irc tasks using the algorithms from Gaussian and at the levels supported by Amesp. *https://github.com/mizu-bai/Gaussian-Amesp.*

Many thanks to the following people for providing the algorithm details documentation and pseudo-code:

Yang Dong :

(1) BFGS, linear search, RFO.

Wang Xubo :

(1) FCI RDM.

(2) Exact two component (X2C) theory.

Bu Yunting :

(1) AIMD.

3. Citation

If Amesp is utilized in your work, the following paper must be cited in your article:

Yingfeng Zhang 2024 Electron. Struct. 6 037002.

DOI: 10.1088/2516-1075/ad5cb5

4. How to use Amesp

4.1 Installation

Amesp is an easy-to-use software, after get the binaries, the user needs to add the name of the directory where the executable files are placed to your PATH environmental variable, for example:

```
export PATH=$PATH:/opt/Amesp/Bin
export KMP_STACKSIZE=4g
ulimit -s unlimited
```

4.2 Running Amesp

To run the Amesp you can use the command as follow:

```
amesp <input> <output>
```

or write amesp <input> directly, then it will create a output file.

5. Features

5.1 Single-point energy calculation

Available methods:

(1) The conventional and RIJ+COSX Hartree-Fock SCF. Include restricted HF (RHF), unrestricted HF (UHF), restricted open-shell HF(ROHF) and HF-3c.

(2) The conventional and RIJ+COSX Kohn–Sham (KS) density functional theory (DFT) . Include restricted KS (RKS), unrestricted KS (UKS), and restricted open-shell KS (ROKS). The functional types include local density approximation (LDA), generalized gradient approximation (GGA), meta-GGA, hybrid generalized gradient approximation (hybrid-GGA), hybrid-meta-GGA, range-separated functionals (RSF), double-hybrid functionals and dispersion corrections.

(3) The conventional and density-fitting MP2, spin-component scaled MP2 (SCS-MP2), scaled opposite-spin MP2 (SOS-MP2), restricted open-shell mp2 (ROMP2). Conventional MP3, spin-component scaled MP3 (SCS-MP3), MP2.5, Conventional MP4.

(4) Configuration-interaction methods and coupled-cluster methods. Include CID, CISD, QCISD, QCISD(T), CCD, CCSD, CCSD(T), CASCI, Full CI.

(5) Multi-configurational Self-Consistent Field Method (MCSCF). Including Complete Active Space Multi-configurational SCF (CASSCF).

(6) Spin-free (SF) exact two-component (X2C) theory for above methods.

(7) Atomic density based tight-binding method (aTB).

(8) Conductor-Like Polarizable-Continuum Model (CPCM) and Universal Easy Solvation Evaluation (uESE) for above methods.

(9) Geometrical counterpoise (gCP) for above methods.

(10) Universal Force Field (UFF), Amber Force Field.

5.2 Electronically excited states

Available methods:

(1) The conventional and RIJ+COSX Configuration interaction singles (CIS) based on RHF, UHF, ROHF.

(2) The conventional and RIJ+COSX time-dependent Hartree-Fock method (TDHF) based on RHF,UHF and ROHF.

(3) The conventional and RIJ+COSX time-dependent Density Functional Theory (TDDFT) for LDA, GGA, meta-GGA, RSF, hybrid-GGA and hybrid-meta-GGA based on RKS,UKS and ROKS.

(4) The conventional and RIJ+COSX Tamm-Dancoff approximation to TDDFT (TDA) for LDA, GGA, meta-GGA, RSF, hybrid-GGA and hybrid-meta-GGA based on RKS,UKS and ROKS.

(5) The conventional and density-fitting(RI) CIS(D) based on RHF ,UHF and TDDFT/TDA for double hybrid-GGA based on RKS ,UKS.

(6) TDDFT-ris and TDA-ris based on R ,U and RO.

(7) Spin-Orbit Coupling calculation for above methods which based on RHF/RKS and ROHF/ROKS.

(8) Non-adiabatic coupling between ground-state and excited-state which based on RHF/RKS and UHF/UKS.

(9) Tamm-Dancoff approximation aTB (TDA-aTB).

(10) Conductor-Like Polarizable-Continuum Model(CPCM) for above methods.

5.3 Geometry optimization

Including minimizations and saddle point optimizations, available methods using analytic gradient :

(1) The conventional and RIJ+COSX Hartree–Fock SCF, Include restricted HF (RHF), unrestricted HF (UHF), restricted open-shell HF(ROHF) and HF-3c.

(2) The conventional and RIJ+COSX Kohn–Sham (KS) density functional theory (DFT) . Include restricted KS (RKS), unrestricted KS (UKS) and restricted open-shell KS(ROKS). The functional types include local density approximation (LDA), generalized gradient approximation (GGA), meta-GGA, hybrid generalized gradient approximation (hybrid-GGA), hybrid-meta-GGA, range-separated functionals (RSF) and dispersion corrections.

(3) The conventional and RI-MP2, spin-component scaled MP2 (SCS-MP2), scaled opposite-spin MP2 (SOS-MP2), and double-hybrid functionals.

(4) The conventional and RIJ+COSX configuration interaction singles (CIS) based on RHF, UHF.

(5) The conventional and RIJ+COSX time-dependent Hartree-Fock method (TDHF) based on RHF,UHF.

(6) The conventional and RIJ+COSX time-dependent Density Functional Theory (TDDFT) for LDA, GGA, RSF and meta-GGA.

(7) The conventional and RIJ+COSX Tamm-Dancoff approximation to TDDFT (TDA) for LDA, GGA, RSF and meta-GGA.

(11) Multi-configurational Self-Consistent Field Method (MCSCF). Including Complete Active Space Multi-configurational SCF(CASSCF).

(12) Spin-free (SF) exact two-component (X2C) theory for above methods.

(13) Atomic density based tight-binding method(aTB).

(11) Tamm-Dancoff approximation aTB (TDA-aTB).

(14) Conductor-Like Polarizable-Continuum Model (CPCM) for above methods.

(15) Geometrical counterpoise (gCP) for above methods.

(16) Universal Force Field(UFF), Amber Force Field.

5.4 Harmonic frequencies

Available methods using analytic hessian :

(1) The conventional and RIJ+COSX Hartree-Fock SCF, Include restricted HF (RHF), unrestricted HF (UHF) and HF-3c.

(2) The conventional and RIJ+COSX Kohn–Sham (KS) density functional theory (DFT) . Include restricted KS (RKS), unrestricted KS (UKS). The functional types include local density approximation (LDA), generalized gradient approximation (GGA), meta-GGA, hybrid generalized gradient approximation (hybrid-GGA), hybrid-meta-GGA, range-separated functionals (RSF) and dispersion corrections.

(3) Atomic density based tight-binding method(aTB).

(4) Conductor-Like Polarizable-Continuum Model(CPCM) for above methods.

(5) Geometrical counterpoise (gCP) for above methods.

(6) Universal Force Field(UFF), Amber Force Field.

Available methods using numerical hessian :

Another methods which have analytic gradient in 5.3.

5.5 Energy Decomposition Analysis

Available methods:

(1) Using Fock matrix.

(2) Mayer's chemical energy component analysis.

5.6 Wavefunction Analysis and Property

Available methods:

(1) Mulliken charges.

(2) Löwdin charges.

(3) Hirshfeld charges.

(4) CM5 charges.

(5) Natural orbital.

(6) Mayer bond order.

(7) Pipek-Mezey and Foster-Boys orbital localization.

(8) Wavefunction stability analysis

(9) Molecular electrostatic potential.

(10) Electric field.

(11) The atomic polarizabilities, dipole moment and quadrupole moment.

5.7 Other methods

Available methods:

(1) Charge transport integral.

(2) Basis set superposition error(BSSE).

(3) Static polarizability and static hyper-polarizability.

(4) Intrinsic Reaction Coordinate(IRC).

(5) Ab Initio Molecular Dynamics(AIMD).

6. Input files

In general, the execution of Amesp is controlled by keywords. The list of the keywords is presented in next section. The options and the corresponding keywords must be given in the input file as the structure of :

```
% maxcore M (MB)
% npara N
! method1 method2 method3 ...
>block1
    keyword11    option11
    keyword12    option12
    ......
```

end

>block2

keyword21 option21

keyword22 option22

.....

end

.....

>xyz charge spin_mult

elem1 coordinate_x1 coordinate_y1 coordinate_z1

elem2 coordinate_x2 coordinate_y2 coordinate_z2

.....

end **//This block must be written.**

Example:

```
! b3lyp def2-SVP RI
>ope
  out 1
  gfout on
end
>scf
  accint 12
end
>xyz 0 1
 H        3.06818648   1.75665122   0.87128909
 H        3.06818648   1.75665122  -0.76170438
end
```

7. Keywords

7.1 Memory and Parallel

Line "**% maxcore M**" specifies the core memory used. For example if you want to set 500MB for Amesp, you need to write "**% maxcore 500**" , for 3GB, you need to write "**% maxcore 3000**" . Default: % maxcore 1000.

Line "**% npara N**" specifies the CPUs used. Default: % npara 1.

Example:

```
% npara 4
% maxcore 2000
!  MP2  cc-pVQZ
>xyz 0 1
  H        3.06818648   1.75665122   0.87128909
  H        3.06818648   1.75665122  -0.76170438
end
```

7.2 Method Line

Line "**! method1 method2 method3...**" specifies the method, basis sets and calculation method. Table1,Table2 and Table3 show the keywords for available methods ,basis sets and calculation method in Amesp. More details for calculation methods are in block **method** in Table7.

Table1. The keywords for available methods in Amesp

method	keyword
Hartree–Fock	hf, hf-3c
LDA	xalpha, svwn5
GGA	pbe, blyp, pw91, bp86, hcth407
meta-GGA	tpss,m06l
Hybrid-GGA	b3lyp, bhlyp, pbe0, b3pw91, b97, b971, b972
Hybird-meta-GGA	tpssh,tpss0,m06-2x,m05-2x
Range-separated functionals	cam-b3lyp, wb97, wb97x, wb97x-d3
Double-hybrid functionals	b2plyp, b2gp-plyp, dsd-pbep86, dsd-blyp
MCSCF	casscf
Møller–Plesset perturbation theory	mp2, mp3, mp4, scs-mp2, sos-mp2, mp2.5, scs-mp3
Configuration-interaction method	cid,cisd,qcisd,qcisd(t),casci,fci
Coupled-cluster method	ccd,ccsd,ccsd(t)
Electronically excited states	cis,cis(d),tdhf
Atomic density based tight-binding method	atb
Force Field	uff, amber, amber94

Note: fci is Full CI

For restricted open-shell situation, you need to add "**ro**" before the keyword, for example: "**rohf**", "**rob3lyp**". And add "**u**" before the keyword, you can use the open-shell method for close-shell system.

Table2. The keywords for available basis sets in Amesp

basis sets	keyword	availability
Pople-style basis sets	sto-2g	H-Xe
	sto-3g	H-Xe
	sto-4g	H-Xe
	sto-6g	H-Xe
	3-21g	H-Xe
	6-31g	H-Kr
	6-31g*/6-31g(d)	H-Kr
	6-31+g*/6-31+g(d)	H-Kr
	6-31g**/6-31g(dp)	H-Kr
	6-31+g**/6-31+g(dp)	H-Kr
	6-31++g**/6-31++g(dp)	H-Kr
	6-311g	H-Kr
	6-311g*/6-311g(d)	H-Kr
	6-311+g*/6-311+g(d)	H-Kr
	6-311g**/6-311g(dp)	H-Kr
	6-311+g**/6-311+g(dp)	H-Kr
	6-311++g**/6-311++g(dp)	H-Kr
	6-31g##	H-Kr
The def/def2 basis sets of the Karlsruhe's group	def2-sv(p)	H-Rn
	def2-svp	H-Rn
	def2-tzvp(-f)	H-Rn

	def2-tzvp	H-Rn
	def2-tzvpp	H-Rn
	def2-qzvp	H-Rn
	def2-qzvpp	H-Rn
	ma-def2-sv(p)	H-Rn
	ma-def2-svp	H-Rn
	ma-def2-tzvp(-f)	H-Rn
	ma-def2-tzvp	H-Rn
	ma-def2-tzvpp	H-Rn
	ma-def2-qzvp	H-Rn
	ma-def2-qzvpp	H-Rn
	def2-svpd	H-La,Hf-Rn
	def2-tzvpd	H-La,Hf-Rn
	def2-tzvppd	H-La,Hf-Rn
	def2-qzvpd	H-La,Hf-Rn
	def2-qzvppd	H-La,Hf-Rn
	def-tzvp	H-Ra
	ma-def-tzvp	H-Ra
Correlation-consistent basis sets	cc-pvdz	H-Ar,Ca-Kr
	cc-pvtz	H-Ar,Ca-Kr
	cc-pvqz	H-Ar,Ca-Kr
	cc-pv5z	H-Ar,Ca-Kr
	aug-cc-pvdz	H-Ar,Sc-Kr
	aug-cc-pvtz	H-Ar,Sc-Kr
	aug-cc-pvqz	H-Ar,Sc-Kr
	aug-cc-pv5z	H-Ar,Sc-Kr
	cc-pvdz-pp	Cu-Kr,Y-Xe,Hf-Rn
	cc-pvtz-pp	Cu-Kr,Y-Xe,Hf-Rn
	cc-pvqz-pp	Cu-Kr,Y-Xe,Hf-Rn

	cc-pv5z-pp aug-cc-pvdz-pp aug-cc-pvtz-pp aug-cc-pvqz-pp aug-cc-pv5z-pp	Cu-Kr,Y-Xe,Hf-Rn Cu-Kr,Y-Xe,Hf-Rn Cu-Kr,Y-Xe,Hf-Rn Cu-Kr,Y-Xe,Hf-Rn Cu-Kr,Y-Xe,Hf-Rn
Miscellaneous and specialized basis sets	lanl2dz	H,Li-La,Hf-Bi
	sdd	H-Rn
	x2c-svpall x2c-sv(p)all x2c-tzvpall x2c-tzvppall	H-Rn H-Rn H-Rn H-Rn
	minao minix	H-Kr H-Rn

Table3. The keywords for available calculation method in Amesp

calculation method	keyword	Note
Excited method	td tda td-ris tda-ris	td tddft td tda td tddft-ris td tda-ris
Density-fitting(RI) and chain of spheres(COSX)	ricosx rimp2	ricosx on ricosx mp2
Geometry optimization	opt optts copt/coptts	opt on opt ts opt on/ts and optcoord car

	zopt/zoptts force numforce	opt on/ts and optcoord zmat force on force num
Harmonic frequencies	freq numfreq	freq on freq num
Intrinsic Reaction Coordinate	irc	irc on
Ab Initio Molecular Dynamics	aimd	aimd on
Static Polarizability and Static Hyper-Polarizability	polar	polar on
Self-Consistent Reaction Field	cpcm uese	scrf cpcm scrf uese
Empirical dispersion corrections	d2 d3 d3bj	dftd d2 dftd d3 dftd d3bj
Symmetry	symm	symm on
X2C	sfx2c1e	x2c sfx2c1e
Spin-Orbit Coupling	soc	soc on
Non-adiabatic coupling	nac	nac on
potential energy surface (PES) scan	scan	scan on
DFT Grid level	grid1 grid2 grid3 grid4 grid5 grid6	grid lv1 grid lv2 grid lv3 grid lv4 grid lv5 grid lv6
COSX Grid level	gridx1 gridx2 gridx3	gridx lv1 gridx lv2 gridx lv3

	gridx4 gridx5	gridx lv4 gridx lv5

A simplest case is presented by using method, basis sets, calculation method and coordinates:

```
! b3lyp  def2-SVP  d3  opt
>xyz 0 1
 C        -0.64814816   0.69023568   0.00000000
 H        -0.29149373  -0.31857432   0.00000000
 H        -0.29147532   1.19463387   0.87365150
 H        -0.29147532   1.19463387  -0.87365150
 H        -1.71814816   0.69024886   0.00000000
end
```

7.3 Basis sets

You can use then basis sets presented in Table2, and Amesp also supports define basis sets for different atoms by the user, by using the keyword "**define**" and block "**>basis**" that under the coordinates you can use the basis sets built-in amesp or reading from the input file. The format of basis sets read from the input file is that used for "**Turbomole**" in the **BSE** library.

The structure of "**>basis**" is :

```
>basis
   elem1   basis1
   elem2   basis2
   ......
end
```

The basis1,basis2 ... in this block must be in Table2 or "**read**","**readecp**", the item of "**read**" means read the basis sets from input file , the "**readecp**" means read the basis sets and ECP sets from input file.

Example:

1: define the basis sets built-in amesp.

```
! HF define
>xyz  0  1
  F          2.54117635    1.14705881    0.00000000
  H          1.94117635    1.14705881    0.00000000
end
>basis
  H  6-31g**
  F  def2-svp
end
```

2: define the basis sets built-in amesp and reading from the input file

```
! HF define
>xyz  0  1
  F          2.54117635    1.14705881    0.00000000
  H          1.94117635    1.14705881    0.00000000
end
>basis
  H  read
  F  def2-svp
end
$basis
*
H    4
*
    2   S
      5.4471780000       0.1562849787
      0.8245472400       0.9046908767
    1   S
      0.1831915800       1.0000000000
*
$end
```

3: define the basis sets reading from the input file

```
! hf define
>xyz 0 1
 S          2.72151879    0.96518986    0.00000000
end
>basis
  S  readecp
end
$basis
 *
 S    1
 *
    2   S
      1.8500000000      -0.5324334973
      0.4035000000       1.2763800937
    1   S
      0.1438000000       1.0000000000
    1   P
      0.1379000000       1.0000000000
 *
$end
$ecp
 *
 S    1
 *
   ncore =  10   lmax =   2
 d-ul      2
   -30.4513290000    2     24.5697664000
   -10.3745886000    2      7.3702438000
 s-ul      2
    93.6460845000    2     15.3706332000
    28.7609065000    2      3.1778402000
 p-ul      2
    53.6569778000    2      7.8120535000
     8.9249559000    2      2.3112730000
 *
$end
```

7.4 Auxiliary sets

When RI or RICOSX are used on SCF or MP2, the auxiliary sets is selected automatically, you can use the block "**>ope**" when you want to choose the auxiliary sets by yourself, please refer to Table 9 for details.

If you choose "define", you can refer to the usage of >basis($basis), just replace the title of the block to >auxscf($auxscf) and >auxmp2($auxmp2).

Table4. Keywords of all auxiliary sets

Keyword	describe	aux sets	availability
auxscf	The aux sets for SCF	def2-j	H-Rn
		def2-jk	H-Rn
		cc-pvtz-jk	H,B-F,Al-Cl,Ga-Br
		cc-pvqz-jk	H,B-F,Al-Cl,Ga-Br

		cc-pv5z-jk	H,B-F,Al-Cl,Ga-Br
auxmp2	The aux sets for MP2	def2-svp-c	H-La,Hf-Rn
		def2-tzvp-c	H-La,Hf-Rn
		def2-tzvpp-c	H-La,Hf-Rn
		def2-tzvpd-c	H-La,Hf-Rn
		def2-tzvppd-c	H-La,Hf-Rn
		def2-qzvpp-c	H-La,Hf-Rn
		def2-qzvppd-c	H-La,Hf-Rn
		cc-pvdz-c	H-Ar,Ga-Kr,Y-Pd,Hf-Pt
		cc-pvtz-c	H-Ar,Sc-Kr,Y-Pd,Hf-Pt
		cc-pvqz-c	H-Ar,Sc-Kr,Y-Pd,Hf-Pt
		cc-pv5z-c	H-Ar,Sc-Kr,Y-Pd,Hf-Pt
		aug-cc-pvdz-c	H-He,B-Ne,Al-Ar,Ga-Kr,Y-Pd,Hf-Pt
		aug-cc-pvtz-c	H-He,B-Ne,Al-Ar,Sc-Kr,Y-Pd,Hf-Pt
		aug-cc-pvqz-c	H-He,B-Ne,Al-Ar,Sc-Kr,Y-Pd,Hf-Pt
		aug-cc-pv5z-c	H-He,B-Ne,Al-Ar,Sc-Kr,Y-Pd,Hf-Pt

Note: (aug-)cc-pvnz-c include (aug-)cc-pvnz-c and (aug-)cc-pvnz-pp-c

7.5 Blocks and options

The blocks start with "**>**" and end with "**end**". For the default case, no blocks need to be present in an input file but they can be present if the user want a detailed control of the program. All blocks are existed in Table 5:

Table5. The blocks in Amesp

scf	Control of the SCF/SCC procedure.
method	Here a calculation method is specified.
posthf	Control of the post-HF procedure.
ope	Control of other operations.
opt	Control of the geometry optimization

freq	Control of the frequency
pcm	Control of the PCM
cas	Control the Complete active space of CASCI and CASSCF
atb	Control of the aTB
irc	Control of the Intrinsic Reaction Coordinate
aimd	Control of the Ab Initio Molecular Dynamics

In block **scf**, sixteen keywords can be used, they were showed in Table6:

Table6. Keywords in block **scf**

Keyword	Default	*Describe* and **options**
maxcyc	125	*The maximum number of SCF/SCC cycles* set to **N** (N≥0)
conver	6	*The SCF/SCC convergence threshold value* RMS Density Matrix change threshold, set to 10^{-N} , and set to $10^{-(N+1)}$ when energy gradient or hessian were required.
accint	10	*2-electron integral and DFT integral accuracy* set to 10^{-N} , and $10^{-(N+1)}$ for RI and COSX
diis	adiis	*Pulay's Direct Inversion in the Iterative Subspace (DIIS) extrapolation method, including CDIIS,ADIIS and EDIIS* **adiis:** Using ADIIS+CDIIS. **ediis:** Using EDIIS+CDIIS. **cdiis:** Using only CDIIS. **qdiis:** Using QDIIS for aTB. **off**: Turn off DIIS.
ndiis	18	*The size of DIIS space* set to **N**
incfock	on	*Incremental Fock method* **off**: Turn off incfock

		on: Turn on incfock
stable	off	*The Wavefunction Stability Analysis* **off**: Turn off Stable **on**: Turn on Stable
soscf	off	*Using Second-Order SCF after energy change less than $10^{-soscfconv}$* **off**: Turn off soscf **1st**: Using the first derivative of Fock matrix and BFGS method. **2nd**: Using the second derivative of Fock matrix and Newton-Raphson method.
soscfconv	2	*The threshold value to turn on SOSCF* Turn on Second-Order SCF after energy change less than $10^{-soscfconv}$
damp	0.7	*The dynamic damping of early SCF iterations* set to **damp** (a real number) D_n+1 = damp*D_n + (1-damp)*D_n+1
fermit	0	*The Fermi Temperature broadening of early SCF iterations* set to **fermit** (a integral number)
vshift	0	*Shift orbital energies by N*0.001 Hartree for SCF iterations* set to **N**
guess	harris	*The initial guess for the HF and DFT wavefunction* *Note: only read works for SCC.* **harris:** Use the harris functional method **huckel:** The extend Huckel method **core:** Diagonalized the core Hamiltonian to form the initial guess **gwh:** The generalized Wolfsberg-Helmholz approximation **read:** The initial guess be read from the mo file **only:** Only calculate initial guess(core) and print
scfmode	auto	*Selects the mode to calculation 2e-integrals* **auto:** According to the input memory to chose direct mode or incore mode **direct:** Choose the direct mode
diistol	0.1	*Turn on the DIIS when the diis error less than diistol* set to **diistol** (a real number)

orbalta orbaltb	0,0 0,0	*Orbit Alter. Swap orbit N1 and Orbit N2 after SCF done.* "orbalta" *for Restrictive Close Shell and Restrictive Open Shell.* "orbalta" *and* "orbaltb" *for Unrestrictive Open Shell.* set to **N1**,**N2**

In block **method**, twenty three keywords can be used, they were showed in Table7:

Table7. Keywords in block **method**

Keyword	Default	*Describe* and **options**
opt	off	*This keyword controls geometry optimization* **on**: Turn on opt for minimizations **ts:** Turin on optts for transition state optimizations **off**: Turn off opt
force	off	*This keyword controls a single calculation of the forces on the nuclei(the gradient of the energy)* **on**: Turn on force **off**: Turn off force **num:** Turn on numerical force
freq	off	*This keyword controls harmonic frequencies* **on**: Turn on freq **off**: Turn off freq **num:** Turn on numerical frequency
irc	off	*This keyword controls IRC* **on**: Turn on IRC **off**: Turn off IRC
aimd	off	*This keyword controls AIMD* **on**: Turn on AIMD **off**: Turn off AIMD
scan	off	*This keyword preforms the rigid or relaxed potential energy surface (PES) scan*

		on: Turn on scan **off**: Turn off scan
scrf	none	*This keyword performs a calculation in the presence of a solvent by placing the solute in a cavity within the solvent reaction field.* **none :** no scrf method be used **cpcm:** using the Conductor-Like Polarizable-Continuum Model(CPCM) **uese:** using the uESE method
ricosx	off	*Using density-fitting (resolution-of-identity) and chain of spheres(COSX) to calculate HF, DFT, MP2* **off**: do not use density-fitting **on**: using RIJ+COSX for HF, DFT and RI-MP2. **mp2**: using density-fitting only for MP2 (including double-hybrid functionals) correlation calculation
td	none	*This keyword controls the excited state calculation using the time-dependent DFT method* **none**: no time-dependent DFT method be used. **tddft**: using time-dependent Density Functional Theory . **tda**: using Tamm-Dancoff approximation to TDDFT. **tddft-ris**: using TDDFT-ris. **tda-ris**: using TDA-ris.
x2c	none	*Using exact two component (X2C) theory* **none :** no x2c method be used **sfx2c1e :** using sf-x2c-1e for one-electronic Hamilton
soc	off	*Calculate the Spin-Orbit Coupling elements between singlet and triplet states based on TDDFT, TDA, TDHF, CIS.* **off**: Turn off SOC **on**: Turn on SOC, only effective for closed-shell systems.
nac	off	*Non-adiabatic coupling matrix elements between ground-state and excited-state* **off**: Turn off nac **on**: Turn on nac, only effective for closed-shell systems and unrestricted open-shell systems
cti	off	*Calculate Charge Transport Integral* *(The rule of the input file is showed in 7.8)* **off**: Turn off cti

		on: Turn on cti
eda	none	*Energy Decomposition Analysis* **none:** do not use EDA **fock:** EDA using fock matrix **mayer**: Mayer's Chemical Energy Component Analysis
polar	off	*Calculate the Static Polarizability and Static Hyper-Polarizability* **on:** Turn on polar **off:** Turn off polar
elefield	off	*Calculate the Electric Field using the SCF density* **on:** Turn on elefield **off:** Turn off elefield
lmo	none	*Localized molecular orbitals* **none**: do not use lmo **pm:** using Pipek-Mezey orbital localization **boys:** using Foster-Boys orbital localization
natorb	none	*Calculate the Natural Orbital* **none**: Turn off natorb **no**: Natural orbital, for R and RO **uno:** unrestricted natural orbital, only for U **nso:** natural spin orbital, only for U
dftd	none	*This keyword controls the calculation of empirical dispersion corrections for DFT and HF calculations using the DFT-D3 approach of Grimme and co-workers* **none**: do not use DFT-D **d2**: calculate the D2 dispersion correction **d3**: calculate the D3 dispersion correction with zero damping **d3bj**: calculate the D3 dispersion correction with BJ damping
field	none 0	*This keyword controls that a finite field be added to a calculation* **none 0**: do not use electric dipole field **x N**: applying an electric dipole field in the X direction of 0.0001*N a.u. **y N**: applying an electric dipole field in the Y direction of 0.0001*N a.u. **z N**: applying an electric dipole field in the Z direction of 0.0001*N a.u.
binding	off	*This keyword controls the binding energy calculation* **bsse:** Turn on binding with BSSE **on:** Turn on binding without BSSE **off:** Turn off binding

atompolar	none	*This keyword controls the Atomic polarizabilities (Tkatchenko-Scheffler), dipole moment and quadrupole moment.* **none:** Turn off atompolar **hirshfeld:** Turn on atompolar with hirshfeld.
gcp	none	*This keyword controls the geometrical counterpoise calculation* **none:** Turn off gcp **hf_minix:** HF/MINIX **hf_def2-svp:** HF/def2-SVP **dft_def2-svp:** DFT/def2-SVP **hf_def2-sv(p):** HF/def2-SV(P) **dft_def2-sv(p):** DFT/def2-SV(P) **hf_6-31g(d):** HF/6-31G(D) **dft_6-31g(d):** DFT/6-31G(D) **hf_minis:** HF/minis **dft_minis:** DFT/minis **hf_def2-tzvp:** HF/def2_TZVP **dft_def2-tzvp:** DFT/def2-TZVP **hf_def-tzvp:** HF/def-TZVP **dft_def-tzvp:** DFT/def-TZVP **hf_cc-pvdz:** HF/cc-pVDZ **dft_cc-pvdz:** DFT/cc-pVDZ **hf_aug-cc-pvdz:** HF/aug-cc-pVDZ **dft_aug-cc-pvdz:** DFT/aug-cc-pVDZ **dft_lanl2dz:** DFT/lanl2dz

In block **posthf**, twelve keywords can be used, they were showed in Table8:

Table8. Keywords in block **posthf**

Keyword	Default	*Describe* and **options**
fc	on	*Using frozen core calculation* **on**: Turn on frozen core **off**: Turn off frozen core
diis	on	*Use of Pulay's Direct Inversion in the Iterative Subspace (DIIS) extrapolation method for CPHF* **on**: Turn on DIIS **off**: Turn off DIIS

ndiis	7	*The size of DIIS space for CPHF* set to **N**
nstates	3	*Solve for N states for CIS, TDHF, TDDFT and TDA* Set to **N**
root	1	*Specifies the "state of interest" for CIS, TDHF, TDDFT and TDA* Set to **N**
tout	1	*Threshold for printing eigenvector components for CIS, TDHF, TDDFT TDA CASCI/CASSCF and Full CI.* set to 10^{-N}
maxcyc	100	*The maximum number of CI/CC and CPSCF cycles* set to **N**
conver	6	*The CI/CC convergence threshold value* energy change, set to 10^{-N}
tdspin	s	*Choice for the spin of excited states, this keyword only effective for closed-shell systems and restricted open-shell systems* **s**: solve for spin=S states. For restrictive close shell situation, this is singlet states **s+1**: solve for spin=S+1 states. For restrictive close shell situation, this is triplet states **all:** solve for spin=S and spin=S+1 states.
tdmode	mo	*integral mode for excited states* **mo:** using MO to calculate excited states, if memory is not enough,it will change to AO mode. **ao**: using AO to calculate excited states.
mdavcyc	30	*The maximum number of davindson iterate cycles for CIS, TDHF, TDDFT, TDA* set to **N**
davconv	4	*The davindson iterate convergence threshold value for CIS, TDHF, TDDFT, TDA* energy change, set to 10^{-N}

In block **ope**, thirteen three keywords can be used, they were showed in Table9:

Table9. Keywords in block **ope**

Keyword	Default	*Describe* and **options**
mofile	on	*Creative the mo (molecular orbital) file* **on**: Turn on mofile **off**: Turn off mofile
trjfile	on	*Creative the trj (trajectory) file with keyword 'scan', 'irc' or 'aimd'* **on**: Turn on trjfile. **off**: Turn off trjfile.
charge	mulliken	*This keyword controls the output the of charge* **mulliken:** Calculate and print Mulliken charge. **lowdin:** Calculate and print Löwdin charge. **hirshfeld:** Calculate and print Hirshfeld charge. **cm5:** Calculate and print CM5 charge. **none:** Do not use charge.
out	0	*This keyword controls the output messages* **0** : Print orbital energy, energy, charge, dipole moment, quadrupole moment, octapole moment. **1** : Print out=0 + Wilson's B-matrix, parts of gradient, parts of Mayer's EDA, UFF/Amber force field informations, aTB element specific parameters. **2** : Print out=1 + molecular orbital, atom distance, density matrix, full mulliken population analysis, gross orbital populations, mayer bond order. **3** : Print out=2 + one-electron integral, γ_{AB}, Γ_{AB} and w_A of aTB. **4** : Print out=3 + AO and MO Two-electron integral.
inttype	sph	*Integral type* **sph**: spherical(5d,7f...) **car**: cartesian(6d,10f...)
gfout	off	*Print current basis sets, ecp sets and auxset* **on**: Turn on gfout **gau**: Print current basis sets, ecp sets and auxset with Gaussian format **off**: Turn off gfout

hessout	on	*Print the Hessian matrix* **on**: Turn on hessout **off**: Turn off hessout
diisout	off	*Print the DIIS coefficients* **on**: Turn on diisout **off**: Turn off diisout
socout	1	*Print the spin-orbit coupling message* **0:** Print EigenValues and EigenVectors of the SOC Matrix. **1:** Print Sum of SOC,EigenValues and EigenVectors of the SOC Matrix. **2:** Print full SOC matrix, Sum of SOC,EigenValues and EigenVectors of the SOC Matrix.
ctiout	1	*Print the Charge Transport Integral message* **1:** Print the orbits only include HOMO, HOMO-1, LUMO, LUMO+1 **2:** Print all orbits
grid	lv3	*DFT grid level* **lv1**: up to 40*350 **lv2**: up to 50*434 **lv3**: up to 60*590 **lv4**: up to 70*770 **lv5**: up to 80*974 **lv6**: up to 90*974
gridx	lv2	*COSX grid level* **lv1**: up to 45*50 **lv2**: up to 55*74 **lv3**: up to 60*86 **lv4**: up to 70*110 **lv5**: up to 75*146
gridfile	off	*Write DFT grid to grid file* **on**: Turn on gridfile

		off: Turn off gridfile
denfile	off	*Write DFT grid and density to den file* **on**: Turn on denfile **off**: Turn off denfile
dffile	off	*Write the dffile for aesp and PFDM* **aesp**: aesp format **pfdm**: pfdm format **off**: Turn off rifile
dftype	eri	*Density fitting type for dffile* **eri:** Using electron repulsion integral for fitting density **ovp:** Using overlap integral for fitting density **antic:** Using anti-Coulomb integral for fitting density
dfset	def2-j	*Set Auxset for dffile and mayerdft* **define:** using user-defined auxset **dfset:** using auxset in Table4
multigrid	off	*Multigrid technology for COSX* **on:** Turn on multigrid **off:** Turn off multigrid
runit	angs	*Specifies the units used for molecular geometries* **angs:** Angstroms will be used **bohr:** Atomic units will be used
symm	off	*Identify the point group of the basis and orbits, note that orbitals are printed in the standard orientation.* **on**: Turn on symm **off**: Turn off symm
symmtol	0.00001	*The threshold value to control the symmetry (bohr)* Set to **symmtol** (a real number)
qphi	keep	*This keyword controls the q*phi and q*q parts of pointcharge*

		keep: Do nothing. **remove:** Remove the q*phi and q*q part.
soctype	zeff	*Way to achieve Spin-Orbital Coupling Hamiltonian* **zeff:** Effective charge approximation
nlmo	occ	*This keyword controls the number of LMO* **occ:** Localize occupied orbitals **vir:** Localize unoccupied orbitals **all:** Localize all orbitals
cosxdamp	0.5	*The damping of COSX overlap fitting* Set to **cosxdamp** (a real number) damp*fitting + (1-damp)*nonefitting
pointcharge	off	*Set background point/gaussian charge be included in the calculation* **on**: Turn on pointcharge using point charge **gau:** Turn on pointcharge using gaussian charge **off**: Turn off pointcharge
gauchgxi	2.0	*The exponent of gaussian charge* Set to **gauchgxi** (a real number) Gauchg(r)=q*(gauchgxi^2/π)^(3/2)*exp(-gauchgxi^2*r^2)
auxscf	default	*Set SCF Auxset* **default:** using default auxset **define:** using user-defined auxset **auxscf:** using auxset in Table4
auxmp2	default	*Set Posh-HF Auxset* **default:** using default auxset **define:** using user-defined auxset **auxmp2:** using auxset in Table4
mayerdft	on	*Calculate the DFT part for Mayer's EDA* **on:** calculate the DFT part **off:** only calculate the HF part

dftdprm	default	*This keyword controls the source of dftd parameters.* **default**: Using the default parameters. **read**: Read the parameters from /Amesp/Lib/Parameter/DFTD3.prm (DFTD2.prm) file.
rsomega	-1.0	*This keyword controls the ω for range-separated functional.* Set to **rsomega** (a real number, rsomega >= 0)

In block **opt**, fourteen keywords can be used, they were showed in Table10:

Table10. Keywords in block **opt**

Keyword	Default	*Describe* and **options**
maxcyc	75	*The maximum number of geom opt cycles* set to **N**
maxstep	0.3	*The maximum size for an optimization step(Bohr)* set to **maxstep** (a real number)
optcoord	red	*Choose the coordinates for optimization* **car:** use cartesian coordinates **red:** use redundant internal coordinates **zmat:** use Z-Matrix coordinates
geom	input	*This keyword specifies the source of the molecule structure* **Input:** molecule structure from the input file **read:** read molecule structure from the mo file
gediis	off	*Use of GEDIIS method for Geometry Optimization* **off:** Turn off GEDIIS **on**: Turn on GEDIIS
ngediis	5	*The size of DIIS space for Geometry Optimization* set to **N**
gediistol	0.001	*Turn on the GEDIIS when the gediis error less than gediistol*

		set to **gediistol** (a real number)
rfo	on	*Rational function optimization* **on:** Turn on RFO **off:** Turn off RFO
scfguess	read	*The scf initial guess in opt iterations* **read:** Change the scf initial guess to “read” **keep:** Keep the scf initial guess unchanged
nr	off	*Using the Newton-Raphson method, this will Calculate the exact Hessian.* **off :** Turn off Newton-Raphson method **on:** Turn on Newton-Raphson method **first:** Only Calculate the exact Hessian on the first cycle
mmopt	auto	*The opt solver for molecular mechanics* **auto:** Select opt solver automatically. **bfgs:** Using BFGS. **cg:** Using conjugate gradient.
inihess	schlegel	*The initial hessian for redundant internal coordinates without “newton on” or “newton first”* **schlegel:** Using schlegel hessian **lindh:** Using lindh hessian **almlof:** Using AlmlÖf hessian **swart:** Using swart hessian **unit:** Using unit matrix
geomtol	normal	*The geom opt convergence threshold value* **loose:** RMS Force = 0.0016667 MAX Force = 0.0025000 RMS Step = 0.0066667 MAX Step = 0.0100000 **normal:** RMS Force = 0.0003000 MAX Force = 0.0004500

		RMS Step = 0.0012000 MAX Step = 0.0018000 **tight:** RMS Force = 0.0000100 MAX Force = 0.0000150 RMS Step = 0.0000400 MAX Step = 0.0000600 **verytight:** RMS Force = 0.0000010 MAX Force = 0.0000015 RMS Step = 0.0000040 MAX Step = 0.0000060
frozen	0 0	*The first and last index of frozen atoms.* set to **index_first index_last**

In block **freq**, four keywords can be used, they were showed in Table11:

Table11. Keywords in block **freq**

Keyword	Default	*Describe* and **options**
step	0.0001	*This keyword controls the step-size for numerical differentiation force and hessian(angstrom,unless runit=bohr has been specified)* Set to **step** (a real number)
scale	1.0	*This keyword controls the frequency scale factor for thermochemistry analysis.* Set to **scale** (a real number)
temper	298.15	*This keyword control the temperature (K)* Set to **temper** (a real number)
press	1.0	*This keyword control the pressure (atm)* Set to **press** (a real number)

In block **pcm**, seven keywords can be used, they were showed in Table12:

Table12. Keywords in block **pcm**

Keyword	Default	*Describe* and **options**
solvent	water	*The solvent and it's dielectric constant in Amesp* **Water :** ε=78.3553 **Acetonitrile:** ε=35.688 **Methanol :** ε=32.613 **Ethanol :** ε=24.852 **Chloroform :** ε=4.7113 **DiethylEther :** ε=4.2400 **Dichloromethane :** ε=8.93 **CarbonTetraChloride :** ε=2.2280 **Benzene :** ε=2.2706 **Toluene :** ε=2.3741 **Acetone :** ε=20.493 **FormicAcid :** ε=51.1 **AceticAcid :** ε=6.2528 **Formamide :** ε=108.94 **TetraHydroFuran :** ε=7.4257 **DiMethylSulfoxide :** ε=46.826 **CycloHexane :** ε=2.0165
eps	-1.0	*User defined dielectric constant* Set to **eps** (a real number, and eps > 10^(-6))
epsinf	1.9	*User defined dynamic (or optical) dielectric constant, only work with eps > 10^(-6).* Set to **epsinf** (a real number)
cpcmx	0.5	*The x for CPCM in f(ε)=(ε-1)/(ε+x).* Set to **cpcmx** (a real number)
cgconv	12	*The convergence threshold value of solving $A^{-1}q$ using CG algorithm* CG threshold, set to 10^{-N}
nleb	lv2	*Number of Lebedev points per atom* **lv1**: 50 **lv2**: 110 **lv3**: 194 **lv4**: 302

pcmout	off	*Print the PCM surface points* **on**: Turn on pcmout. **off**: Turn off pcmout.

In block **cas**, eleven keywords can be used, they were showed in Table13:

Table13. Keywords in block **cas**

Keyword	Default	*Describe* and **options**
cas	0 0	*The Complete active space* set to **N M** (N electrons and M orbitals)
nroots	1	*Set the roots for CASSCF, CASCI, FCI* Set to **N**
iroot	0	*Specifies the "state of interest" for CASSCF* *The iroot was defaulted to iroot=nroots if iroot=0* Set to **i**
sa	on	*Prefer a State-Averaged CASSCF calculation* **on**: Turn on sa. **off**: Turn off sa.
fcidump	off	*Print fcidump and rdm to fci file and rdm file* **on**: Turn on fcidump. **off**: Turn off fcidump.
guess	harris	*The initial guess for the CASSCF wavefunction* **harris**: Use the harris functional method **hf**: The Hartree-Fock's wavefunciton **read**: The initial guess be read from the mo file **core**: Diagonalized the core Hamiltonian to form the initial guess
maxcyc	100	*The maximum number of CASSCF cycles* set to **N**
conver	6	*The CASSCF Energy convergence threshold value*

		Energy change threshold, set to 10^{-N}
gconv	4	*The CASSCF Grad convergence threshold value* Grad change threshold, set to 10^{-N}
mdavcyc	15	*The maximum number of davindson iterate cycles for CASCI/FCI* set to **N**
davconv	6	*The davindson iterate convergence threshold value for CASCI/FCI* energy change, set to 10^{-N}

In block **atb**, three keywords can be used, they were showed in Table14:

Table14. Keywords in block **atb**

Keyword	Default	*Describe* and **options**
gammalevel	2nd	*This keyword controls the level of gamma.* **2nd**: The second order gammma. **3rd**: The third order gamma. **3rddiag**: The third order diagonal gamma.
atbprm	default	*This keyword controls the source of atb parameters.* **default**: Using the default parameters. **read**: Read the parameters from /Amesp/Lib/Parameter/aTB.prm file.
etdamax	-1.0	*The excited energy cutoff for TDA-aTB(eV).* Set to a real number(unit in eV), if etdamax > 0, using excited energy cutoff and full diagonalization, if etdamax < 0 ,using davidson diagonalization.

In block **irc**, three keywords can be used, they were showed in Table15:

Table15. Keywords in block **irc**

Keyword	Default	*Describe* and **options**
ircsolver	lqa	*This keyword controls the solver of IRC.* **lqa**: Use the local quadratic approximation(LQA) method. **euler**: Use the Euler method, this option is not recommended.

maxpoints	10	*This keyword controls the maxpoints of IRC.* set to **maxpoints** (a integer)
maxstep	0.1	*This keyword controls the maxstep of IRC.* set to **maxstep** (a real number, unit bohr)

In block **aimd**, six keywords can be used, they were showed in Table16:

Table16. Keywords in block **aimd**

Keyword	Default	*Describe* and **options**
dt	1.0	*This keyword controls the delta time (fs) of AIMD* set to **dt** (a real number)
maxsteps	100	*This keyword controls the maxsteps of AIMD.* set to **maxsteps** (a integer)
temper	298.15	*This keyword control the temperature (K) of AIMD* Set to **temper** (a real number)
press	1.0	*This keyword control the pressure (atm) of AIMD* Set to **press** (a real number)
relaxtime	50.0	*This keyword controls the relaxation time (fs) of AIMD* set to **relaxtime** (a real number)
ensemble	nvt	*This keyword controls the ensemble.* **nvt**: The NVT ensemble. **nve**: The NVE ensemble.

7.6 Coordinates

The input of molecular coordinates support two types: Cartesian and Z-matrix. For Cartesian, you need to write the block keyword "**>xyz**",and "**zmat**" for Z-matrix.

Example:

Cartesian

```
% npara 4
!  DSD-PBEP86  def2-TZVP
>scf
  conver  10
end
>xyz 0 1
 N        2.73484737   1.28525105   0.05479235
 H        3.06816926   0.34243796   0.05479235
 H        3.06818648   1.75665122   0.87128909
 H        3.06818648   1.75665122  -0.76170438
end
```

Z-matrix

```
! mp2 def2-TZVP
>zmat  0 1
 C
 F      1     1.070
 F      1     1.070     2     109.471
 F      1     1.070     2     109.471     3      120.00
 I      1     1.350     2     109.471     3     -120.00
end
```

The Dummy Atoms can be used in Z-matrices,which are represented by the pseudo atomic symbol X:

```
! HF def2-SVP
>zmat 0 1
  X
  C     1    1.8
  O     2    R2     1   90.0
  O     2    R2     1   90.0    3  -180.00
end
>coord
   R2    1.5
end
```

You can use the mechanics type Bq (e.g., O-Bq) to set a ghost atom, for example:

```
% npara 4
!  b3lyp  def2-TZVP
>xyz 0 2
 N        2.73484737   1.28525105   0.05479235
 H-Bq     3.06816926   0.34243796   0.05479235
 H        3.06818648   1.75665122   0.87128909
 H        3.06818648   1.75665122  -0.76170438
 N-Bq     4.06818648   1.75665122  -0.76170438
end
```

7.7 Scan

This performs the potential energy surface (PES) scan, Table7 shows how to use it, and it supports rigid/relaxed PES scan and less than 6 variables now, the structure of input file for rigid PES scan is :

```
! method1 method2 method3 ...
>zmat  charge  spin_mult
  elem1
  elem2   con1   var1
  elem2   con2   var2   con3   var3
  ......
end
>coord
  var1   coord1
  var2   coord2
  ......
end
>scan
  var1   NSteps1   Step1
  var2   NSteps2   Step2
  ......
end
```

The molecular structure must be defined using Z-matrix coordinates for PES scan. For relaxed PES scan, one just use zmat optimization, a simple way is to add “zopt” to the method line.

Example:

```
% npara 4
! HF  def2-SVP  scan
>zmat 0 1
 C
 H    1    R1
 H    1    R1    2    A1
 H    1    R1    2    A1    3    D1
 H    1    R1    2    A1    3   -120.0
end
>coord
  R1   1.00
  A1   108.0
  D1   120.0
end
>scan
  R1   2   0.5
  A1   1   2.0
end
```

This input causes variable R1 to be stepped 2 times by 0.5. Thus, three R1 values (1.0, 1.5 and 2.0) will be done for each combination of other variables. Similarly, 2 values for A1 (120.0, and 122.0) will be used, and D1 will be held fixed at 120.0.

7.8 Group

This block is used for Binding and Charge Transport Integral,it divided the molecular into serious groups, the structure of block >group is :

```
>group
    iatmbegin1   iatmend1   charge1   spin_mult1
    Iatmbegin2   iatmend2   charge2   spin_mult2
    Iatmbegin3   iatmend3   charge3   spin_mult3
   ........
end
```

Example of Charge Transport Integral :

```
! b3lyp def2-TZVP
>method
  cti  on
end
>xyz 0 1
 C          -2.45886640   0.00914077   0.00000000
 H          -2.10221198  -0.99966923   0.00000000
 H          -2.10219356   0.51353896   0.87365150
 H          -2.10219356   0.51353896  -0.87365150
 H          -3.52886640   0.00915395   0.00000000
 O          -0.90493596  -0.24680073   0.00000000
 H           0.05506404  -0.24680073   0.00000000
 H          -1.22539055   0.65813510   0.00000000
end
>group
  1 5 0 1
  6 8 0 1
end
```

Example of Binding :

```
! b3lyp def2-TZVP
>method
  binding bsse
end
>xyz 0 1
 C          -2.45886640   0.00914077   0.00000000
 H          -2.10221198  -0.99966923   0.00000000
 H          -2.10219356   0.51353896   0.87365150
 H          -2.10219356   0.51353896  -0.87365150
 H          -3.52886640   0.00915395   0.00000000
 O          -0.90493596  -0.24680073   0.00000000
 H           0.05506404  -0.24680073   0.00000000
 H          -1.22539055   0.65813510   0.00000000
end
>group
  1 5 0 1
  6 8 0 1
end
```

7.9 Point charge

This block is used for setting point charge, the structure of block >charge is :

```
>charge
      x1     y1     z1    charge1
      x2     y2     z2    charge2
     ........
end
```

Example of point charge:

```
! b3lyp def2-SVP
>ope
  pointcharge on
end
>xyz 0 1
 C          1.56370654  -0.42471042   0.00000000
 H          1.92036097  -1.43352042   0.00000000
 H          1.92037938   0.07968777   0.87365150
 H          1.92037938   0.07968777  -0.87365150
 H          0.49370654  -0.42469723   0.00000000
end
>charge
  2.0   2.0   2.0   1.2
  2.0  -2.0   2.0   1.1
end
```

7.10 Zeff

This block is used for setting zeff for SOC, the structure of block >zeff is :

```
>zeff
    elem1 zeff1
    elem2 zeff2
    ........
end
```

Example of zeff:

```
! B3LYP 3-21G TD SOC
>xyz 0 1
  N        1.91455682   1.24999998   0.00000000
  H        2.24787872   0.30718689   0.00000000
  H        2.24789593   1.72140016   0.81649673
  H        2.24789593   1.72140016  -0.81649673
end
>zeff
  N   5.0
end
```

7.11 Mass

This block is used for setting atom mass, the structure of block >mass is :

```
>mass
    Atom_index1    mass1
    Atom_index2    mass2
    ........
end
```

Example of mass:

```
% npara 4
! hf 3-21g opt freq
>xyz 0 1
 O          2.34423690    0.91900310    0.00000000
 H          2.02378232    1.82393893    0.00000000
 Cl         3.99423690    0.91900310    0.00000000
end
>mass
  2   2.0141
end
```

7.12 MMCharge

This block is used for setting UFF and Amber charge, the structure of block >mmcharge is :

```
>mmcharge
      Atom_index1    charge1
      Atom_index2    charge2
     ........
end
```

Example of mmcharge:

```
% npara 4
! uff
>xyz 0 1
 C          0.68988175    0.75558475    0.00000000
 C          2.01579775    0.75558475    0.00000000
 H          0.09629675    1.67962275    0.00000000
 H          0.09626575   -0.16842925   -0.00002200
 H          2.60938275   -0.16845325   -0.00001900
 H          2.60941375    1.67959875    0.00002600
end
>mmcharge
  3   0.1
  5  -0.1
end
```

8. Other Programs

amg : Transform amesp input file to gaussian input file or transform gaussian input file to amesp input file.

qcfc : Transform gaussian input file to other program's input file.

m2m : Transform mo file to molden file.

m2w : Transform mo file to wfn file.

m2a : Transform mo file to amesp's mo file(.amo).

z2c : Transform the input file with Z-matrix to Cartesian.

c2z : Transform the input file with Cartesian to Z-matrix.

o2g : Get the final geometry from aop file and write to gjf file.

aesp : amesp's esp program, it calculate esp and rho.

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